MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

PUNE, INDIA
A Leap Towards World Class Education

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Assignment - 1.

Q1. Interpreted language - program are executed line by line, which makes debugging easier High Level language - No need to Manage memory explicity: python handles it automatically open Source - Free to use & distribute Extensive library Support - Coms with a large standard library for tasks like file handling web development.

- advantages

1. fast devllopment - less code is required compared to other language like C or Java.

2. Easy debugging & maintanance - Simple syntax reduces & makes program easier to maintain.

3. Versatile: Used in web devllopment, machine learning, AI, data science, automation & more.

any two python IDES

- Python
- Jupyter notebook

2. History & Evolution.

- Python was created by Guido Van Rossum in 1989.
- Released in 1991
- Named after Monty python
- Designed for Simplicity & readibility
- Python 3x is the latest & widely used version.

Why python is popular

- Easy Syntax for beginners
- Used in AI, Ml, web development, Data science.
- Huge Community support
- used by companies like google, Netflix, Insta.

3.
```
a = float (input ("Enter first number: "))
b = float (input ("Enter second number: "))

print ("Addition", a+b)
print ("Substraction=" a-b)
print ("Multiplication, "a*b)
print ("Division" = a/b)
```

output.
Enter first number: 40
enter second number: 5
addition 45
substraction 35

Multiplication = 50
division = 20

4. a = 10
b = 5.5
c = 3+4j
d = True
print(a+b, type(c), d)

Data types
- a → int
- b → float
- c → complex
- d → bool

output explanation.
- a+b = 15.5
- type(c) → <class 'complex'>
- d → true

5 Relational operators : Used for comparison, used is conditional & decision making.
a = 10
b = 5
print (a>b)

- logical operators : used to combine conditions used in authentication & validation

x = true
y = false
print (x & y)

- Bitwise operators: operate on binary numbers
  a = 5
  b = 3
  print (a & b)

6. - accepts user input
   - performs arithmetic, relational, logic & assignment operators.
   - Display result clearly.

   The program should handle int, float & boolean data types.

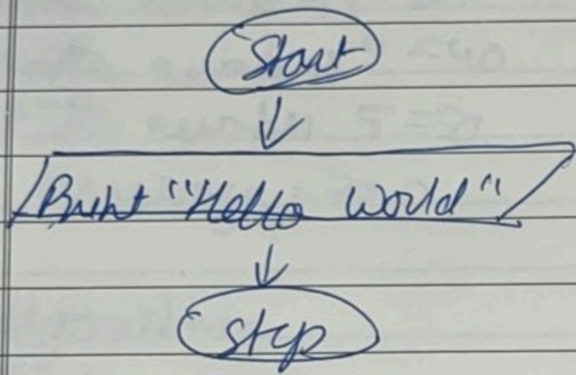
```
a = int (input ("Enter first number :"))
b = int (input ("Enter second number :"))

print ("1. Arithmatic")
print ("2. Relational")
print ("3. logical")
print ("4. assignment")

ch = int (input ("Enter choice :"))
if ch == 1
    print ("Add = ", a+b)
elif
    ch = 2
  print ("a>b = ", a>b)
```

```
ch = 4:
  a+ = b
  print("a=", a)
else
    print ("invalid choice")
```

7.



Code. print ("Hello world")

Output
Hello world

Algorithm
2. Start
2. Print Hello world
3. Stop

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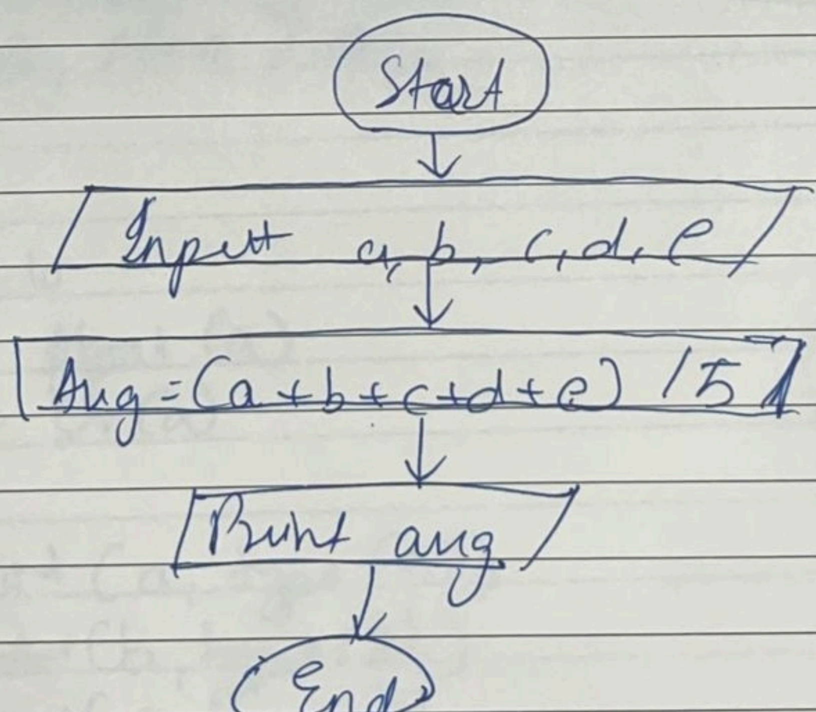
```
a = float (input ("Enter number 1:"))
b = float (input ("Enter number 2:"))
c = float (input ("Enter number 3:"))
d = float (input ("Enter number 4:"))
e = float (input ("Enter number 5:"))
```

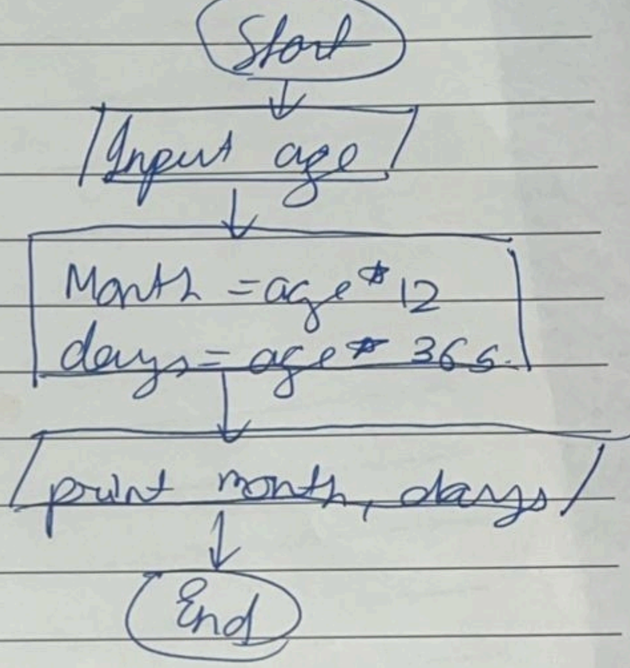
avg = (a+b+c+d+e)/5
print("average =", avg)
output.
Enter number 1=10
enter number 2=20
enter number3 =30
enter number 4 =40
enter number 5 =50
average = 30.0

Algorithm.
1. Start
2. Accept 5 numbers from user
3. Calculate average
4. Display average
5. Stop.



Q9
```
age=int(input("Enter age in years:"))
months=age*12
days=age*365
print("Age in months:", months)
print("Age in days:", days)
```

Output
Enter age in years: 18
Age in months : 216
age in days : 6570



• algorithm
1. Start
2. Input age in years
3. Calculate Months = age*12
4. Calculate day = age*365
5. Display Month & days
6. stop

10.
```
a = 10
b = float(a)
c = str(a)

print(a, type(a))
print(b, type(b))
print(c, type(c))
```

output
10 <class 'int'>
10.0 <class 'float'>
10 <class 'str'>

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Algorithm
1. Start
2. Assign integer value
3. Convert integer to float & string
4. Display values & their data types
5. Stop.